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ORIGINAL PAPER

Optimal operation and valuation of electricity storage systems in intraday markets

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Abstract
This paper applies computational techniques of convex stochastic optimization to optimal operation and valuation of electricity storage systems in the face of uncertain electricity prices. Our valuations are based on the indifference pricing principle, which builds on optimal trading strategies and calibrates to the user's financial position, market views and risk preferences. The underlying optimization problem is solved with the Stochastic Dual Dynamic Programming algorithm, which is applicable to various specifications of storage systems, and it allows for e.g. hard constraints on storage capacity and charging speed. We illustrate the approach in intraday trading where the agent charges or discharges a battery over a finite number of delivery periods, and the electricity prices are subject to bid-ask spreads and significant uncertainty. Optimal strategies are found in a matter of minutes on a regular PC. We find that the corresponding trading strategies and battery valuations vary consistently with respect to the agent's risk preferences as well as the physical characteristics of the battery.

1 Introduction

The growing proportion of renewable energy generation increases the uncertainties and seasonalities of supply and the price of electricity. This creates financial incentives to use batteries and other electricity storage systems for trading electricity in order to benefit from the price fluctuations. It can be expected that when the overall storage capacity increases and such activities become more popular, the aggregate supply and price fluctuations will diminish. Further discussion and references on the

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topic can be found e.g. in Yu and Foggo (2017) and the survey article Weitzel and Glock (2018).

This article proposes mathematical models and computational tools for optimal operation and valuation of batteries and other electricity storage systems in the context of electricity trading. Economically sensible valuation and optimization techniques will facilitate the expansion of the overall storage capacity, which will be necessary for the development of resilient power systems based on renewables. The presented techniques optimize storage management strategies in the face of uncertain electricity prices, and they adapt to the trader's financial position, risk preferences as well as different market conditions. Besides the simple model specifications studied in this paper, the techniques can be applied to various specifications of the storage as well as to extensions where the agent might have existing storage, production facilities or consumption or a mixture of the above.

We start by developing a stochastic optimization model for finding optimal strategies for charging and discharging an energy storage over time,subject to physical constraints. The aim is to optimize a user-specified risk measure of the terminal wealth, which depends on the uncertain price fluctuations as well as on the decision maker's strategy. Realistic models of the storage result in stochastic optimization problems that fail to be analytically solvable. This is partly due to the fact that a physical storage has capacity constraint,s which amount to state constraints in the corresponding optimal control formulation. We will employ the *stochastic dual dynamic programming* algorithm which is able to handle convex instances of such problems computationally in cases where the underlying risk factors are Markovian; see Dowson and Kapelevich (2021) and its references.

Being able to solve the optimal storage management problem numerically allows us to compute *indifference prices* for batteries and other storage systems. Indifference pricing is a general approach for determining economically meaningful prices of (financial) products whose cashflows cannot be replicated by trading liquid financial instruments; see e.g. the collection (Carmona 2009) and the references there. In the numerical illustrations, we study an intraday trading problem where an agent manages a battery by charging or discharging it over 24 delivery periods, and the trading takes place in the intraday market. We show, in particular, how the physical characteristics of the storage and the trader's risk preferences affect the valuations.

Optimal management of power systems under uncertainty has been recently surveyed in Roald et al. (2023), with the stochastic dual dynamic programming (SDDP) algorithm identified as one of the most promising methods for the management of energy storage systems (Löhndorf et al. 2013; Mégel et al. 2015; Löhndorf and Wozabal 2021; Zheng et al. 2022). As opposed to alternatives such as model predictive control (see e.g. Arnold and Andersson (2011)), SDDP takes explicitly into account the future uncertainties when optimizing decisions. Continuous-time approaches based on solving the Hamilton-Jacobi-Bellman equations, on the other hand, struggle with state constrain,ts which are necessary for realistic descriptions of storage syste,ms which usually involve hard constraints on the amount of stored energy. The recent paper (Picarelli and Vargiolu 2021) studies the optimal operation of a pumped hydroelectric storage with state constraints. The authors assume positive electricity prices and linear utility funct,ions which do not support general risk

preferences. They treat state constraints by the "level set approach", where nondifferentiable penalty functions are added to the objective. Similar techniques were applied by Germain et al. (2023) in the context of mean field games with some applications to renewable energy. Closer to the present paper is Löhndorf and Wozabal (2021) which studied gas storage valuation using the SDDP algorithm. As opposed to that of Löhndorf and Wozabal (2021), our model allows for lending or borrowing money in a cash account/money market. This avoids "discounting" future cashflows and it leads to a more natural notion of the indifference price of a storage in terms of money paid at the time of purchase. Moreover, thanks to the explicit modeling of the agent's cash position, we avoid the delicate "asset-backed trading strategies" of Löhndorf and Wozabal (2021) that, at a given point t in time, involve planned future decisions in addition to the decisions to be implemented at time t.

In the context of electricity markets, indifference pricing has been studied in Porchet et al. (2009) and Callegaro et al. (2017) in the continuous-time setting. The first one applied indifference pricing to production facilities that are controlled by switching between a finite number of states. The second one studied the pricing of a general class of structured contracts. A particularly interesting example there is a "virtual storage" contract, which resembles a physical storage in terms of the generated cashflows. The numerical computations in Callegaro et al. (2017), however, are based on finite difference approximations of the associated Hamilton-Jacobi-Bellman (HJB), which do not adapt well to state constraints.

The rest of this paper is organized as follows: Sect. 2 starts with a formulation of the battery management problem as a stochastic optimization problem. The problem is then used to define the indifference price of a battery. The section concludes with a simple model of the electricity prices observed at the end of the intraday market for each delivery period. Section 3 reformulation of the storage management problem as an instance of a standard convex control problem. Section 4 gives a short review of the dynamic programming theory and the Stochastic Dual Dynamic Programming algorithm for the general problem formulation from Sect. 3. Section 5 gives the computational results for a battery management problem over 24 delivery periods.

2 Battery management in the intraday market

Consider the problem of managing a battery by charging or discharging it over a finite number T of delivery periods. Before 30 September 2025, there were 24 one-hour delivery periods in a day, while after that date, this was changed to 96 15-minute delivery periods. At the beginning of each delivery period, the trader decides on the amount of energy bought or sold during the delivery period. The energy is used to charge or discharge the battery. The agent aims to maximize the total revenue generated by trading over the T periods.

Section 2.1 below gives a brief description of the intraday electricity market, and Sect. 2.2 formulates the optimization problem that will be studied in the remainder of the paper. Section 2.3 recalls the notion of indifference price and discusses its numerical computation. Section 2.4 describes the stochastic model that will be used to describe electricity prices in the computational experiments in Sect. 5.

2.1 The intraday market

During each delivery period, energy can be bought or sold at the *imbalance prices*; see RTE (2024). The imbalance prices are published a couple of minutes after the end of the delivery period, so they remain unknown during the delivery period when the physical electricity trading occurs. Electricity can be traded also before the delivery period in the *intraday market*, which is essentially a futures market for the physical commodities of energy delivered over the delivery periods. Each delivery period is associated with one futures market. Much like exchange-traded securities, the intraday market is based on the *double auction* mechanism where traders submit *limit orders* in continuous time. All unmatched orders are recoded in the *limit order book* that gives the marginal futures prices for buying electricity in a given delivery period.

At any given time t, the limit order book for a given delivery period $t' \geq t$ can be represented by a nondecreasing *marginal price curve* $s_{t,t'} : \mathbb{R} \to \mathbb{R}$ so that the cost of buying u units of energy is given by

$$S_{t,t'}(u) := \int_0^u s_{t,t'}(v)dv; \tag{1}$$

see e.g., [Pennanen (2011), Section 2]. Since $s_{t,t'}$ is nondecreasing, the cost function $S_{t,t'}$ is convex, and the best *bid* and *ask prices* are given by its left- and right-derivatives at the origin:

$$s^-_{t,t'} = \lim_{h \nearrow 0} \frac{S_{t,t'}(u)}{u} = \lim_{u \nearrow 0} s_{t,t'}(u) \quad \text{and} \quad s^+_{t,t'} = \lim_{h \searrow 0} \frac{S_{t,t'}(u)}{u} = \lim_{u \searrow 0} s_{t,t'}(u); \tag{2}$$

see [Rockafellar (1970), Theorem 24.2]. We always have $s^-_{t,t'} < s^+_{t,t'}$, since otherwise, the best bid and ask offers would have been matched and taken out of the limit order book.

2.2 The optimization problem

When trading at the imbalance prices, one has to make trading decisions before knowing the associated prices. To avoid such a price risk, one can trade in the intraday markets where the futures prices for each delivery period are observable in real time. This gives rise to a continuous-time portfolio optimization problem where one manages a portfolio of futures contracts, one for each delivery period. Such high-dimensional portfolio optimization problems are, in general, difficult to solve. Moreover, a mathematical formulation of such a problem would require a stochastic model for the high-dimensional forward curve that gives the futures prices for each delivery period. Even if one was able to solve the associated optimization problem, one would be faced with the "model risk" that comes with the specification of such high-dimensional stochastic models.

We take the intermediate approach where energy for the tth delivery period is traded in the intraday market, but only just before the closing of the corresponding futures market. For example, we could observe the limit order book for the tth

delivery period a small unit ϵ of time (e.g. 5 seconds) before the intraday market for that period closes. We denote the corresponding cost function by $S_t := S_{t-\epsilon,t}$; see Eq. (1). Unlike the imbalance prices, the cost function S_t is known at time t when making the decision to trade energy for the tth delivery period.

We study a simple storage with a maximum capacity of $\overline{X}^e$ and maximum charging and discharging speeds of $\overline{U}$ and $\underline{U}$, respectively. If u units of energy is input to the storage, the charging level increases by $c(u)$ units. The function $c : \mathbb{R} \to \mathbb{R}$ is strictly increasing and satisfies $c(u) \leq u$ for every $u \in \mathbb{R}$. Again, negative u corresponds to discharging the battery. If x units of energy is stored at the beginning of period $[t, t+1[$, the charging level will be $(1-l)x$ units at the end of the period. The constant $l \in [0, 1]$ describes proportional storage loss per period.

To formulate the battery management problem mathematically, we describe the uncertainties and information by a probability space $(\Omega, \mathcal{F}, P)$ and a filtration $(\mathcal{F}_t)_{t=0}^T$, respectively, and assume that each cost function S_t is an $\mathcal{F}_t$-measurable convex *normal integrand*[1]. The problem can then be written as

$$\begin{aligned} \text{maximize} \quad & \mathbb{E}[v(X_T^m)] \quad \text{over} \quad (X^m, X^e, U) \in \mathcal{N} \\ \text{subject to} \quad & X_0^m = x_0^m, \\ & X_0^e = 0, \\ & X_t^m = X_{t-1}^m - S_t(U_t) \quad t = 1, \dots, T, \\ & X_t^e = (1-l)X_{t-1}^e + c(U_t) \quad t = 1, \dots, T, \\ & X_t^e \in [0, \overline{X}^e] \quad t = 1, \dots, T, \\ & U_t \in [-\underline{U}, \overline{U}] \quad t = 1, \dots, T, \end{aligned} \tag{3}$$

where X_t^m is the cash position and X_t^e is the amount of stored energy, respectively, at time t. The control variable U_t specifies the amount of energy bought at time t. Here and in what follows, $\mathcal{N}$ denotes the space of *nonanticipative* (i.e. $(\mathcal{F}_t)_{t=0}^T$-adapted) strategies. The constraint $(X^m, X^e, U) \in \mathcal{N}$ thus means that decisions taken at time t can only depend on the information observed by time t. As usual, negative purchases are interpreted as sales. The initial wealth x_0^m is fixed. The *utility function* $v : \mathbb{R} \to \mathbb{R}$ is assumed increasing and concave.

2.3 Indifference pricing

Indifference pricing is a general principle for defining economically meaningful values of (financial) products in *incomplete markets* where prices cannot be determined by arbitrage arguments alone; see e.g. Carmona (2009) and Pennanen (2014) and their references. In general, indifference prices are *subjective* in the sense that they depend on an agent's existing financial position, risk preferences and views concerning the future development of the underlying risk factors (as described by their

[1] Recall that a function $f : \mathbb{R}^n \times \Omega \to \overline{\mathbb{R}}$ is an $\mathcal{F}_t$-measurable *normal integrand* if the epigraphical mapping $\omega \mapsto \operatorname{epi} f(\cdot, \omega)$ is a closed-valued and $\mathcal{F}_t$-measurable; see e.g. [Rockafellar and Wets (1998), Chapter 14].

probabilistic model). For replicable claims, however, the indifference prices coincide with risk-neutral valuations; see [Pennanen (2014), Theorem 4.1].

We will study an agent who considers renting a battery for a fixed time period $[0, T]$. The *indifference price* for renting the battery is the maximum amount the agent could pay for it without worsening their financial position. In the context of problem (3), the indifference price is given by

$$\pi := \sup\{p \in \mathbb{R} | \varphi(x_0^m - p, \overline{X}^e) \geq \varphi(x_0^m, 0)\}, \tag{4}$$

where, for any $x, X \in \mathbb{R}$, $\varphi(x, X)$ denotes the optimum value of (3) when $x_0^m = x$ and $\overline{X}^e = X$. Thus, $\varphi(x_0^m - p, \overline{X}^e)$ is the highest achievable expected utility when the agent pays p units of cash in exchange for getting access to a battery with maximum capacity $\overline{X}^e$. The value $\varphi(x_0^m, 0)$, on the other hand, is the highest achievable expected utility for the agent if they decide not to rent the battery. The indifference price is thus the highest sensible price for an agent who measures their wellbeing by the expected utility of terminal wealth as in problem (3).

Note that, in this simple model where the agent either manages the battery or does nothing at all, we simply have $\varphi(x_0^m, 0) = v(x_0^m)$. The indifference principle can equally well be applied in situations where the agent already owns storage and is considering renting more of it. Similarly, one could study situations where the agent has consumption or production facilities or any mixture of the above. In model (3), we have assumed that all revenue is invested in cash, but one could extend the model also by allowing investments in alternative assets. The model would then involve additional decision variables that describe the investment strategy. The indifference pricing principle applies equally well in such more complicated situations.

If the function $x \mapsto \varphi(x, U)$ is continuous and strictly increasing, the indifference price π is the unique solution to the equation

$$\varphi(x_0^m - \pi, \overline{X}^e) = \varphi(x_0^m, 0) \tag{5}$$

provided a solution exists. Finding a solution of the equation (5) is a nontrivial problem in general since a single evaluation of the function φ requires the numerical solution of problem (3). The task simplifies, however, in the case of the exponential utility function

$$v(z) = \frac{1}{\rho}[1 - \exp(-\rho z)], \tag{6}$$

where $\rho > 0$ is a given risk aversion parameter.

Proposition 1 *In the case of the exponential utility function (6), the indifference price (4) is given by*

$$\pi = -\frac{\ln(1 - \rho\varphi(0, \overline{X}^e))}{\rho}.$$

Proof The change of variables $X_t^m \to X_t^m - x$ in problem (3) shows that the optimum value $\varphi(x_0^m, \overline{X}^e)$ is not affected if we set the initial wealth x_0^m to zero and add x_0^m units of cash to the terminal wealth X_T^m instead. Thus, in the case of exponential utility, the optimum value function satisfies

$$\varphi(x, U) = e^{-\rho x}\varphi(0, U) + \frac{1}{\rho}(1 - e^{-\rho x}).$$

Equation (5) can thus be written as

$$e^{-\rho(x_0^m - \pi)}\varphi(0, \overline{X}^e) + \frac{1}{\rho}(1 - e^{-\rho(x_0^m - \pi)}) = e^{-\rho x_0^m}\varphi(0, 0) + \frac{1}{\rho}(1 - e^{-\rho x_0^m})$$

or equivalently

$$e^{\rho\pi}\varphi(0, \overline{X}^e) - \frac{1}{\rho}e^{\rho\pi} = \varphi(0, 0) - \frac{1}{\rho}.$$

This has the unique solution

$$\pi = \frac{\ln(1 - \rho\varphi(0, 0)) - \ln(1 - \rho\varphi(0, \overline{X}^e))}{\rho},$$

where $\varphi(0, 0) = 0$. □

Proposition 1 implies that the indifference price of the battery in (4) can be found by solving only one instance of problem (3). In problems where the agent has e.g. production facilities, the computation of the indifference price would require the solution of two optimization problems since, in that case, the computation of the optimum value $\varphi(0, 0)$ obtained without renting the battery becomes nontrivial, unlike in the proof above.

2.4 The electricity price model

In the numerical illustrations in Sect. 5 below, we assume both S_t and c to be positively homogeneous in the decision variables. More specifically, we will assume that

$$S_t(u, \omega) = \begin{cases} s_t^+(\omega)u & \text{if } u \geq 0, \\ s_t^-(\omega)u & \text{if } u \leq 0, \end{cases} \tag{7}$$

where s_t^+ and s_t^- denote the best ask and the best bid prices, respectively, just before the tth delivery period; see (2). The bid and ask prices are modeled as stochastic processes. The above assumption can be justified for a small trader whose trades are unlikely to exceed the liquidity available at the best bid and ask prices. Larger orders may dig deeper into the limit order book so that the marginal prices would move

against the trader according to equation (1). Simple stochastic models for the full limit order book can be found e.g. in Malo and Pennanen (2012).

For simplicity, we will assume that $s_t^- = s_t - \delta$ and $s_t^+ = s_t + \delta$, where $\delta > 0$ is constant and only the *mid-price* s_t is stochastic. We will assume that

$$s_t = \bar{s}_t + \xi_t,$$

where ξ_t is a zero-mean stochastic process and $\bar{s}_t$ is the *day-ahead* price for the tth delivery period observed after the day-ahead auction. The logic is that, being futures prices for energy in the coming delivery periods, the day-ahead prices should be close to the expected intraday prices of the corresponding delivery periods. Consistent positive or negative deviations from day-ahead prices would allow for simple "statistical arbitrage" trades, so one would expect the prices to converge according to supply and demand. Figure 1 plots the historical time series of day-ahead and ID1 prices for France in 2024 at hourly granularity. The plot in the bottom of Fig. 1 shows the difference of the two series.

The ID1 index is a weighted average of executed trades during the last hour before the closure of the intraday market. We take it as a proxy for the intraday mid-price in our statistical model below.

We will describe the process $\xi = (\xi_t)$ by the autoregressive time series model

$$\xi_t = a\xi_{t-1} + \epsilon_t \tag{8}$$

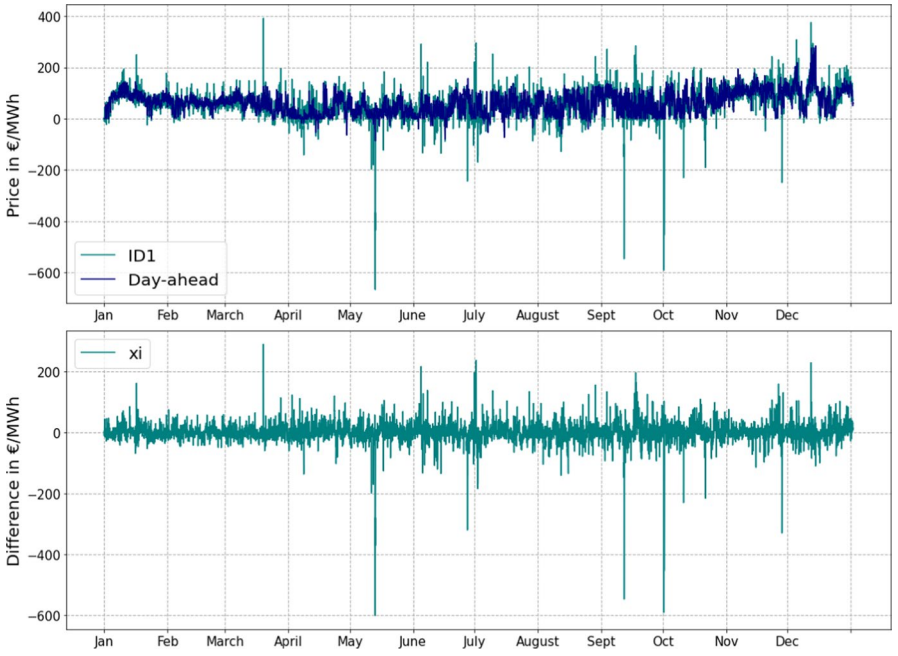


Fig. 1 Time series of day-ahead and ID1 prices (the top plot) and their difference (bottom plot) for France in 2024

where ϵ_t are iid Gaussians with zero mean. We estimate the coefficient $a \in \mathbb{R}$ with a linear regression from 8760 observations of ξ_t (365 days of hourly data). Table 1 summarizes the results of the linear regression. The high p-value of the intercept supports the assumption that ξ has zero mean. We will assume that the random term ϵ_t in (8) follows the zero-mean Gaussian distribution. Its variance is estimated from the regression residuals

$$r_t = \xi_{t+1} - \hat{a}\xi_t.$$

3 Convex control reformulations

This section reformulates the storage management problem (3) as a *convex stochastic optimal control* problem. For generality, we will allow for the general control format of [Carpentier et al. (2023), Section 6.2] where, at each stage t, the system may be controlled by both $\mathcal{F}_{t-1}$-measurable and $\mathcal{F}_t$-measurable controls; see also Pennanen and Perkkiö (2026). Such formulations arise naturally, e.g. when trading in both intraday and imbalance markets. Other instances can be found in Parra et al. (2024).

Consider the optimal control problem

$$\text{minimize} \quad \mathbb{E}\left[\sum_{t=0}^{T-1} L_t(X_t, U_t^b, U_{t+1}^a) + J(X_T)\right] \text{ over } \quad (X, U^b, U^a) \in \mathcal{N}, \tag{9a}$$

$$\text{subject to } X_0 = x_0, \tag{9b}$$

$$X_{t+1} = F_t(X_t, U_t^b, U_{t+1}^a) \quad t = 0, \dots, T-1 \ a.s., \tag{9c}$$

where the state X_t and the controls[2] U_t^b and U_t^a are random variables taking values in $\mathbb{R}^N$, $\mathbb{R}^{M^b}$ and $\mathbb{R}^{M^a}$. The functions $L_t : \mathbb{R}^N \times \mathbb{R}^{M^b} \times \mathbb{R}^{M^a} \times \Omega \to \overline{\mathbb{R}}$ and $J : \mathbb{R}^N \times \Omega \to \overline{\mathbb{R}}$ are extended real-valued convex normal integrands and, for each t,

$$F_t(x_t, U_t^b, U_{t+1}^a, \omega) := A_{t+1}(\omega)x_t + B_{t+1}^b(\omega)U_t^b + B_{t+1}^a(\omega)U_{t+1}^a + W_{t+1}(\omega),$$

where A_{t+1}, B_{t+1}^b and B_{t+1}^a are $\mathcal{F}_{t+1}$-measurable random matrices of appropriate dimensions and W_{t+1} are $\mathcal{F}_{t+1}$-measurable random vectors. As before, the set $\mathcal{N}$

Table 1 Results of the linear regression (8)

	Value	p-value
Slope $\hat{a}$	0.48	0.00
Intercept	−0.13	0.65
R^2	0.23	.

[2] The superscript b in U_t^b stands for "before" (in the sense of a decision taken at the beginning of the time interval $[t, t+1[$), whereas the superscript a in U_t^a stands for "after" (in the sense of a decision taken at the end of the time interval $[t, t+1[$).

denotes the space of adapted stochastic processes. The vector $x_0 \in \mathbb{R}^N$ is a given initial state.

The format of problem (9) is quite general and can accommodate various kinds of storage systems and market specifications. Besides storage management and electricity trading, it can include production facilities and/or consumption of electricity. The computational techniques discussed below apply to such situations as well. Both production and consumption may create stronger incentives to buy storage systems. The indifference pricing approach described in Sect. 2.3 applies directly to such modifications.

Problem (3) does not fit the format of (9), in general, since the functions S_t and c are nonlinear. However, when the cost functions are of the form (7), we can rewrite the budget constraint in (3) as the linear equation

$$X_t^m = X_{t-1}^m - s_t^+ U_t^+ + s_t^- U_t^-,$$

where U_t^+ and U_t^- are the positive and negative parts, respectively, of the control variable U_t. As to the battery, we will assume that the function c is of the form

$$c(u) = \begin{cases} c^+ u & \text{if } u \geq 0, \\ c^- u & \text{if } u \leq 0 \end{cases} \tag{10}$$

where c^+ and c^- are constants such that $0 < c^+ \leq c^-$. We can then write the battery constraint in the linear form

$$X_t^e = (1-l)X_{t-1}^e + c^+ U_t^+ - c^- U_t^-.$$

When the cost function is given by Equation (7) and the function c is given by Equation (10), Problem (3) can be written with *linear* system equations as

$$\begin{aligned} \text{maximize} \quad & \mathbb{E}[v(X_T^m)] \quad \text{over} \quad (X^m, X^e, U) \in \mathcal{N} \\ \text{subject to} \quad & X_0^m = x_0^m, \\ & X_0^e = 0, \\ & X_t^m = X_{t-1}^m - s_t^+ U_t^+ + s_t^- U_t^- \quad t = 1, \ldots, T, \end{aligned} \tag{11a}$$

$$X_t^e = (1-l)X_{t-1}^e + c^+ U_t^+ - c^- U_t^- \quad t = 1, \ldots, T, \tag{11b}$$

$$X_t^e \in [0, \overline{X}^e] \quad t = 0, \ldots, T, \tag{11c}$$

$$U_t^+ \in [0, \overline{U}] \quad t = 0, \ldots, T, \tag{11d}$$

$$U_t^- \in [0, \underline{U}] \quad t = 0, \ldots, T\,. \tag{11e}$$

Consider also the relaxation

$$\begin{array}{ll} \text{maximize} & \mathbb{E}[v(X_T^m)] \quad \text{over} \quad (X^m, X^e, \hat{U}, \check{U}) \in \mathcal{N} \\ \text{subject to} & X_0^m = x_0^m, \\ & X_0^e = 0, \\ & X_t^m = X_{t-1}^m - s_t^+ \hat{U}_t + s_t^- \check{U}_t \quad t = 1, \ldots, T, \end{array} \tag{12a}$$

$$X_t^e = (1-l)X_{t-1}^e + c^+ \hat{U}_t - c^- \check{U}_t \quad t = 1, \ldots, T, \tag{12b}$$

$$X_t^e \in [0, \overline{X}^e] \quad t = 0, \ldots, T, \tag{12c}$$

$$\hat{U}_t \in [0, \overline{U}] \quad t = 0, \ldots, T, \tag{12d}$$

$$\check{U}_t \in [0, \underline{U}] \quad t = 0, \ldots, T, \tag{12e}$$

where we allow $\hat{U}$ and $\check{U}$ to take strictly positive values at the same time. Problem 12 is a *convex* stochastic control problem. If we add the complementarity constraint that $\hat{U}_t \check{U}_t = 0$ for every t, then Problem (12) is equivalent to Problem (11). Thus, Problem (12) is a relaxation of Problem (11) and thus, its optimum value of (12) is at least as high as that of (11). Adding the complementarity constraint would, however, destroy convexity and make the problem difficult to solve, in general. The following Proposition 2 gives a simple condition under which the complementarity condition is redundant in the sense that, the optimum values in problems (12) and (11) coincide, and the optimal solutions of problem (11) are easily constructed from the solutions of (12).

Proposition 2 *Assume that*

$$\frac{s_t^-}{c^-} \leq \frac{s_t^+}{c^+}. \tag{13}$$

Then the optimum values of (11) and (12) coincide. Moreover, if $(X^m, X^e, \hat{U}, \check{U})$ *is feasible in (12), and we define* $(\bar{X}^m, \bar{X}^e, \bar{U})$ *by setting*

$$C_t := c^+ \hat{U}_t - c^- \check{U}_t \tag{14}$$

and

$$\bar{U}_t := C_t^+ / c^+ - C_t^- / c^-, \tag{15a}$$

$$\bar{X}_t^m := \bar{X}_{t-1}^m - s_t^+ \bar{U}_t^+ + s_t^- U_t^-, \quad \bar{X}_0^m = x_0^m, \tag{15b}$$

$$\bar{X}_t^e := X_t^e, \quad \bar{X}_0^e = 0, \tag{15c}$$

then $(\bar{X}, \bar{U})$ *is feasible in (11) and*

$$\mathbb{E}[v(\bar{X}_T^m)] \geq \mathbb{E}[v(X_T^m)]. \tag{16}$$

Proof As already pointed out, the optimum value of Problem (11) is dominated by that of Problem (12). To prove the converse, it suffices to prove the second claim of statement.

Let $(X^m, X^e, \hat{U}, \check{U})$ be feasible in Problem (12), and define $(\bar{X}^m, \bar{X}^e, \bar{U})$ according to (14) and (15a)-(15c). We first show that $(\bar{X}^m, \bar{X}^e, \bar{U})$ is feasible in Problem (11). For each $t = 1, \ldots, T$, we have to consider two cases according to the sign of C_t but we just consider the case $C_t \geq 0$ as the case $C_t \leq 0$ is treated in a similar way. If $C_t \geq 0$, then $\bar{U}_t \geq 0$ and thus,

$$\bar{U}_t^- = 0 \quad \text{and} \quad 0 \leq \bar{U}_t^+ = \bar{U}_t = \hat{U}_t - \frac{c^-}{c^+}\check{U}_t \leq \hat{U}_t \leq \overline{U}\ , \tag{17}$$

where the last inequality holds since $\hat{U}_t$ satisfies (12d). By (17), $\bar{U}$ satisfies constraints (11d) and (11e). Since $C_t \geq 0$, we get

$$\begin{aligned} s^+\bar{U}_t^+ - s^-\bar{U}_t^- &= s^+\bar{U}_t && (\text{as } \bar{U}_t^+ = \bar{U}_t \text{ and } \bar{U}_t^- = 0 \text{ when } \bar{U}_t \geq 0) \\ &= s^+\left(\hat{U}_t - \frac{c^-}{c^+}\check{U}_t\right) && (\text{by (17)}) \\ &\leq s^+\hat{U}_t - s^-\check{U}_t \end{aligned} \tag{18}$$

where the last inequality holds by assumption (13). On the other hand,

$$\begin{aligned} c^+\bar{U}_t^+ - c^-\bar{U}_t^- &= c^+(C_t/c^+) && (\text{by (15a) as } C_t^+ = C_t \text{ and } C_t^- = 0 \text{ when } C_t \geq 0) \\ &= c_t^+\hat{U}_t - c_t^-\check{U}_t\ , \end{aligned} \tag{19}$$

where the second equality holds by (14). Proceeding in a similar way when $C_t \leq 0$ shows that (18) and (19) holds for arbitrary C_t.

Equation (19) together with the initial conditions implies that X^e and $\bar{X}^e$ coincide and thus, $\bar{X}^e$ satisfies (11c) since X^e satisfies (12c). Thus, we have proved that $(\bar{X}^m, \bar{X}^e, \bar{U})$, is feasible in Problem (11).

It remains to prove (16). Since the utility function v is nondecreasing, it suffices show that $X_T^m \leq \bar{X}_T^m$. We use induction on t. Clearly, $X_0^m = x_0^m = \bar{X}_0^m$. Assume now that that $X_{t-1}^m \leq \bar{X}_{t-1}^m$. We get

$$\begin{aligned} X_t^m &= X_{t-1}^m - \left(s_t^+\hat{U}_t - s_t^-\check{U}_t\right) && (\text{by (12a)}) \\ &\leq \bar{X}_{t-1}^m - \left(s_t^+\hat{U}_t - s_t^-\check{U}_t\right) && (\text{by the induction assumption}) \\ &\leq \bar{X}_{t-1}^m - \left(s^+\bar{U}_t^+ - s^-\bar{U}_t^-\right) && (\text{by (18)}) \\ &= \bar{X}_t^m, && (\text{by (11a)}) \end{aligned}$$

so, by induction, $X_t^m \leq \bar{X}_t^m$ for all $t = 0, \ldots, T$. □

Problem (12) is in the format of (9) without the here-and-now controls U^b. Indeed, problem (12) is an instance of (9) with $N = 2$, $M^b = 0$, $M^a = 2$,

$$J(x_T) = -\upsilon(x_T^m),$$

$U_t^a = (\check{U}_t, \hat{U}_t)$ and

$$L_t(X_t, U_t^b, U_{t+1}^a) = \begin{cases} 0 & \text{if } X_t^e \in [0, \overline{X}^e], \hat{U}_{t+1} \in [0, \overline{U}] \text{ and } \check{U}_{t+1} \in [0, \underline{U}], \\ +\infty & \text{otherwise.} \end{cases}$$

We can thus use the numerical techniques described in Sects. 4.2 and 4.3 below. In particular, since there are no here-and-now controls, we can use the SDDP Julia package described in Dowson and Kapelevich (2021).

4 Stochastic dual dynamic programming

In general, problem (9) does not allow for analytical solutions, so we will resort to computational techniques based on dynamic programming. Section 4.1 reviews the general dynamic programming theory for optimal control problems of the form (9). In particular, Theorem 3 characterizes optimal solutions in terms of Bellman functions (aka cost-to-go functions) while Theorem 5 shows that when the underlying randomness is driven by a Markov process, the Bellman functions only depend on the current system state X_t and the current value of the Markov process. Section 4.2 presents the discretization of Markov processes. Section 4.3 describes a version of the *stochastic dual dynamic programming* algorithm for problems of the form (9).

4.1 Stochastic dynamic programming

We will assume throughout that there is a stochastic process $\xi = (\xi_t)_{t=0}^T$ such that L_{t-1}, A_t, B_t^b, B_t^a and W_t in () depend on ω only through $\xi^t := (\xi_s)_{s=0}^t$ and J_T depends on ω only through ξ^T. We will also assume that ξ_t takes values in $\mathbb{R}^{d_t}$ and we denote

$$d^t := d_0 + \cdots + d_t$$

and $d = d^T$. Without loss of generality, we will assume that $\Omega = \mathbb{R}^d$ and that $\xi_t(\omega)$ is the projection of ω to its tth component and that $\mathcal{F}_t$ is the σ-algebra on $\mathbb{R}^d$ generated by the projection $\xi \mapsto \xi^t := (\xi_0, \ldots, \xi_t)$. In particular, P is the distribution of ξ. With a slight misuse of notation we will write $L_t(x_t, U_t^b, U_{t+1}^a, \xi^{t+1})$ instead of $L_t(x_t, U_t^b, U_{t+1}^a, \omega)$ and similarly for other $\mathcal{F}_{t+1}$-measurable random elements.

It follows that the $\mathcal{F}_t$-conditional expectation of an $\mathcal{F}_{t+1}$-measurable quasi-integrable random variable $\xi \mapsto \eta(\xi^{t+1})$ has a pointwise representation

$$\mathbb{E}\big[\eta(\xi^{t+1})|\xi^t\big] := \int_{\mathbb{R}^{d_{t+1}}} \eta(\xi^t, \xi_{t+1}) d\nu_t(\xi^t, d\xi_{t+1}) \quad \forall \xi^t \in \mathbb{R}^{d^t},$$

where ν_t is a *regular* ξ^t*-conditional distribution of* ξ_{t+1}; see, e.g., [Pennanen and Perkkiö (2024), Section 2.1.5]. Note that $\xi^{t+1} = (\xi^t, \xi_{t+1})$. We say that a function $h : \mathbb{R}^n \times \mathbb{R}^{d^{t+1}} \to \overline{\mathbb{R}}$ is *lower bounded* if there is an integrable function $m : \mathbb{R}^{d^{t+1}} \to \mathbb{R}$ such that $h(x, \xi^{t+1}) \geq m(\xi^{t+1})$ for all $(x, \xi^{t+1}) \in \mathbb{R}^n \times \mathbb{R}^{d^{t+1}}$. Given a lower bounded measurable function $h : \mathbb{R}^n \times \mathbb{R}^{d^{t+1}} \to \overline{\mathbb{R}}$, we denote

$$\mathbb{E}[h(x, \xi^{t+1})|\xi^t] := \int_{\Xi_{t+1}} h(x, \xi^t, \xi_{t+1}) d\nu_t(\xi^t, d\xi_{t+1}) \quad \forall (x, \xi^t) \in \mathbb{R}^n \times \mathbb{R}^{d^t}.$$

We say that a sequence of lower-bounded measurable functions $J_t : \mathbb{R}^N \times \mathbb{R}^{d^t} \to \overline{\mathbb{R}}$ solves the Bellman equations for () if $J_T = J$ and

$$J_t(x_t, \xi^t) = \inf_{U_t^b} \left\{ \mathbb{E}\left[\inf_{U_{t+1}^a} \{L_t(x_t, U_t^b, U_{t+1}^a, \xi^{t+1}) + J_{t+1}(F_t(x_t, U_t^b, U_{t+1}^a, \xi^{t+1}), \xi^{t+1})\} \middle| \xi^t \right] \right\} \quad (20)$$

for all $t = 0, \ldots, T-1$ and P-almost every ξ. Bellman equations of the above form (20), together with optimal feedbacks, can be derived[3] from the framework of [Carpentier et al. (2023), Section 6] (and Carpentier et al. (2023), [Section 3.3] for optimal feedbacks), which first analyzed Bellman equations in the case of nonlinear system dynamics with decision-hazard-decision information structures. The following result is from [Pennanen and Perkkiö (2026), [Theorem 1].

Theorem 3 (Optimality principle) *If* $(J_t)_{t=0}^T$ *is a solution of (20) such that each* $J_t : \mathbb{R}^N \times \Xi^t \to \overline{\mathbb{R}}$ *is a lower bounded convex normal integrand, then the optimum value of () equals* $J_0(x_0)$ *and an* $(\bar{X}, \bar{U}^b, \bar{U}^a) \in \mathcal{N}$ *solves () if and only if* $\bar{X}_0 = x_0$ *and*

$$\bar{U}_t^b(\xi^t) \in \operatorname*{argmin}_{U_t^b \in \mathbb{R}^{M^b}} \left\{ \mathbb{E}\left[\inf_{U_{t+1}^a \in \mathbb{R}^{M^a}} \{L_t(\bar{X}_t, U_t^b, U_{t+1}^a, \xi^{t+1}) + J_{t+1}(F_t(\bar{X}_t, U_t^b, U_{t+1}^a, \xi^{t+1}), \xi^{t+1})\} \middle| \xi^t \right] \right\},$$

$$\bar{U}_{t+1}^a(\xi^{t+1}) \in \operatorname*{argmin}_{U_{t+1}^a \in \mathbb{R}^{M^a}} \{L_t(\bar{X}_t(\xi^t), \bar{U}_t^b(\xi^t), U_{t+1}^a, \xi^{t+1}) + J_{t+1}(F_t(\bar{X}_t(\xi^t), \bar{U}_t^b(\xi^t), U_{t+1}^a, \xi^{t+1}), \xi^{t+1})\},$$

$$\bar{X}_{t+1}(\xi^{t+1}) = F_t(\bar{X}_t(\xi^t), \bar{U}_t^b(\xi^t), \bar{U}_{t+1}^a(\xi^{t+1}), \xi^{t+1})$$

for all $t = 0, \ldots, T-1$ *and P-almost every* ξ.

The next result gives sufficient conditions for the existence of solutions to the Bellman equations (20). Given a normal integrand $h : \mathbb{R}^n \times \Omega \to \overline{\mathbb{R}}$, the function h^∞,

[3] In [Carpentier et al. (2023), Sect. 6], Proposition 13 can be extended to the case where instantaneous costs, dynamics and conditional distribution of the next uncertainty all three depend upon the whole sequence of past uncertainties (and not only upon the last uncertainty). The proof requires an adaptation of Appendix D in the preprint version of Carpentier et al. (2023) (which contains additional material compared to the published version (Carpentier et al. 2023)). We can then obtain a variant of [Carpentier et al. (2023), Equation (42)] that corresponds to Equation (20) above. Then, optimal feedbacks can be deduced from [Carpentier et al. (2023), Subsect. 3.3].

obtained from h by defining $h^\infty(\cdot,\omega)$, for each $\omega\in\Omega$, as the *recession function* of $h(\cdot,\omega)$, is a positively homogeneous convex normal integrand; see e.g. Pennanen and Perkkiö (2024) [Example 14.54a] or Pennanen and Perkkiö (2024) [Theorem 1.39]. The following is Pennanen and Perkkiö (2026) [Theorem 2] with the first stage running cost redefined as $+\infty$ unless $X_0=x_0$.

Theorem 4 (Existence of solutions) *Assume that the set*

$$\Big\{(X,U^b,U^a)\in\mathcal{N}\,\Big|\,\sum_{t=0}^{T-1}L_t^\infty(X_t,U_t^b,U_{t+1}^a)+J^\infty(X_T)\le 0,$$
$$X_0=0\ ,\ \ X_t=A_tX_{t-1}+B_t^bU_{t-1}^b+B_t^aU_t^a\ ,\ \ t=1,\dots,T\ a.s.\Big\}$$

is linear. Then the control problem () has an optimal solution and the associated Bellman equations (20) have a unique solution $(J_t)_{t=0}^T$ of lower-bounded convex normal integrands.

The following is from Pennanen and Perkkiö (2026) [Theorem 3].

Theorem 5 (Markovianity) *Let $(J_t)_{t=0}^T$ be a sequence of lower-bounded normal integrands that solve (20). Assume that $\xi=(\xi_t)_{t=0}^T$ is a Markov process and that L_{t-1}, A_t, B_t^b, B_t^a and W_t depend on ξ only throug (ξ^t,ξ^{t-1}). Then J_t depends on ξ^t only through ξ_t. If ξ is a sequence of independent random variables and if L_{t-1}, A_t, B_t^b, B_t^a and W_t do not depend on ξ^{t-1}, then J_t do not depend on ξ at all.*

4.2 Discretization of Markov processes

In general, the Bellman equations (20) do not allow for analytical solutions. We will therefore resort to numerical approximations and assume, as in Theorem 5, that there is a Markov process $\xi=(\xi_t)_{t=0}^T$ such that L_{t-1}, A_t, B_t and W_t depend on ξ only through (ξ_{t-1},ξ_t). We will first approximate the Markov process ξ by a finite-state Markov chain, and then use the Stochastic Dual Dynamic Programming (SDDP) algorithm to solve the discretized problem; see Sect. 4.3 below.

We will discretize ξ using integration quadratures as in Tauchen and Hussey (1991) in the context of option pricing. To that end, we assume that the ξ_t-conditional distribution of ξ_{t+1} has density $p_t(\cdot|\xi_t)$. Given a strictly positive probability density ϕ_{t+1} on $\mathbb{R}^{d_{t+1}}$, the conditional expectation of a random variable $\xi\mapsto\eta(\xi^{t+1})$ will be approximated by

$$\begin{aligned}\mathbb{E}[\eta|\xi_t]&=\int_{\mathbb{R}^{d_{t+1}}}\eta(\xi_{t+1})p_t(\xi_{t+1}|\xi_t)d\xi_{t+1}\\&=\int_{\mathbb{R}^{d_{t+1}}}\eta(\xi_{t+1})\frac{p_t(\xi_{t+1}|\xi_t)}{\phi_{t+1}(\xi_{t+1})}\phi_{t+1}(\xi_{t+1})d\xi_{t+1}\\&\approx\sum_{i=1}^N\eta(\xi_{t+1}^i)\frac{p_t(\xi_{t+1}^i|\xi_t)}{\phi_{t+1}(\xi_{t+1}^i)}w_{t+1}^i,\end{aligned}$$

where $(\xi_{t+1}^i, w_{t+1}^i)_{i=1}^N$ is a quadrature approximation of the density ϕ_t. A natural choice for ϕ_{t+1} would be the unconditional density of ξ_{t+1} provided it is available in closed form. Constructing such approximations for $t = 0, \dots, T$ and conditioning each ξ_{t+1} with the quadrature points ξ_t^i obtained at time t results in a finite state Markov chain where, at each t, the process $(\xi_t)_{t=0}^T$ takes values in the finite set $\{\xi_t^j\}_{j=1}^N$ and the transition probability from ξ_t^j to ξ_{t+1}^i is given by

$$p_t^{ji} := \frac{p_t(\xi_{t+1}^i | \xi_t^j)}{\phi_{t+1}(\xi_{t+1}^i)} w_{t+1}^i.$$

Sufficient conditions for the weak convergence of such discretizations are discussed in Karr (1975).

4.3 Stochastic dual dynamic programming algorithm

The stochastic dual dynamic programming (SDDP) algorithm is a computational technique for solution of Bellman equations in convex stochastic optimization problems governed by finite-state Markov chains. The algorithm is based on successive approximation of the cost-to-go functions $(J_t)_{t=0}^T$ by polyhedral lower approximations. The SDDP algorithm was introduced in Pereira and Pinto (October 1991) for stagewise independent processes ξ and later extended to finite-state Markov processes ξ in Infanger and Morton (1996); see also Philpott and de Matos (2012); Löhndorf and Shapiro (2019) and the references there. The following version of the SDDP algorithm from Pennanen and Perkkiö (2026) applies to control problems of form (). Note that, when the underlying process ξ is a finite-state Markov chain, the conditional expectations below are finite sums weighted by the transition probabilities.

0. **Initialization:** Choose convex (polyhedral) lower-approximations J_t^0 of the cost-to-go functions J_t and set $k = 0$.
1. **Forward pass:** Sample a path ξ^k of ξ and define X^k by $X_0^k = x_0$ and

$$U_t^{bk} \in \operatorname*{argmin}_{U_t^b \in \mathbb{R}^{M^b}} \mathbb{E}\left[\inf_{U_{t+1}^a \in \mathbb{R}^{M^a}} \{L_t(X_t^k, U_t^b, U_{t+1}^a) + J_{t+1}^k(F_t(X_t^k, U_t^b, U_{t+1}^a))\} \middle| \xi_t^k \right],$$

$$U_{t+1}^{ak} \in \operatorname*{argmin}_{U_{t+1}^a \in \mathbb{R}^{M^a}} \{L_t(X_t^k, U_t^{bk}, U_{t+1}^a, \xi_t^k, \xi_{t+1}^k) + J_{t+1}^k(F_t(X_t^k, U_t^{bk}, U_{t+1}^a, \xi_t^k, \xi_{t+1}^k), \xi_{t+1}^k)\},$$

$$X_{t+1}^k = F_t(X_t^k, U_t^{bk}, U_{t+1}^{ak}, \xi_t^k, \xi_{t+1}^k).$$

2. **Backward pass:** Let $J_T^{k+1} := J$ and, for $t = T-1, \dots, 0$,

$$\tilde{J}_t^{k+1}(X_t^k, \xi_t^k) := \inf_{U_t^b \in \mathbb{R}^{M^b}} \mathbb{E}\left[\inf_{U_{t+1}^a \in \mathbb{R}^{M^a}} \{L_t(X_t^k, U_t^b, U_{t+1}^a) + J_{t+1}^{k+1}(F_t(X_t^k, U_t^b, U_{t+1}^a))\} \middle| \xi_t^k \right],$$

$$V_t^{k+1} \in \partial \tilde{J}_t^{k+1}(X_t^k, \xi_t^k)$$

and let

$$J_t^{k+1}(x_t, \xi_t^k) := \max\{J_t^k(x_t, \xi_t^k), \tilde{J}_t^{k+1}(X_t^k, \xi_t^k) + V_t^{k+1} \cdot (x_t - X_t^k)\} \quad \forall x_t \in \mathbb{R}^N.$$

Set $k := k + 1$ and go to 1.

Above, the dot " " stands for the Euclidean inner product. The minimization problems in the above algorithm are two-stage stochastic optimization problems, while the original problem (9) has T decision stages. Note also that the third minimization in the forward pass is one of the second-stage problems already solved in the second minimization. The functions J_t^k generated by the algorithm are polyhedral lower approximations of the true cost-to-go functions J_t. It follows that the minimization problems in the above recursions are finite-dimensional optimization problems that can be solved with standard convex optimization solvers. If the functions L_t and J are polyhedral, the problems are LP problems.

In the decision-observation format, where only the control U^b is present, the two-stage problems reduce to a single-stage stochastic optimization problem, while in the observation-decision format without the U^b control, only deterministic single-stage problem need to be solved. This version of the algorithm is implemented e.g. in the Julia package described in Dowson and Kapelevich (2021).

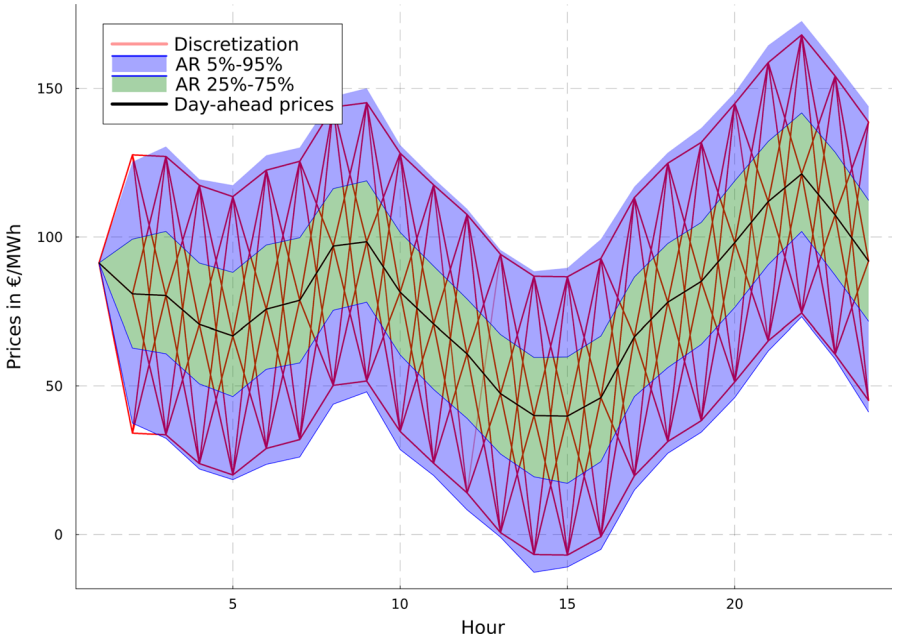


Fig. 2 AR-1 process and its discretization as a Markov chain with 3 quadrature points per period

Table 2 Time to compute 1000 iterations of SDDP as we increase the number N of quadrature points in the discretization of the AR process

N	1	2	4	8	16	32	64
Time (s)	201	44	77	144	270	820	1820

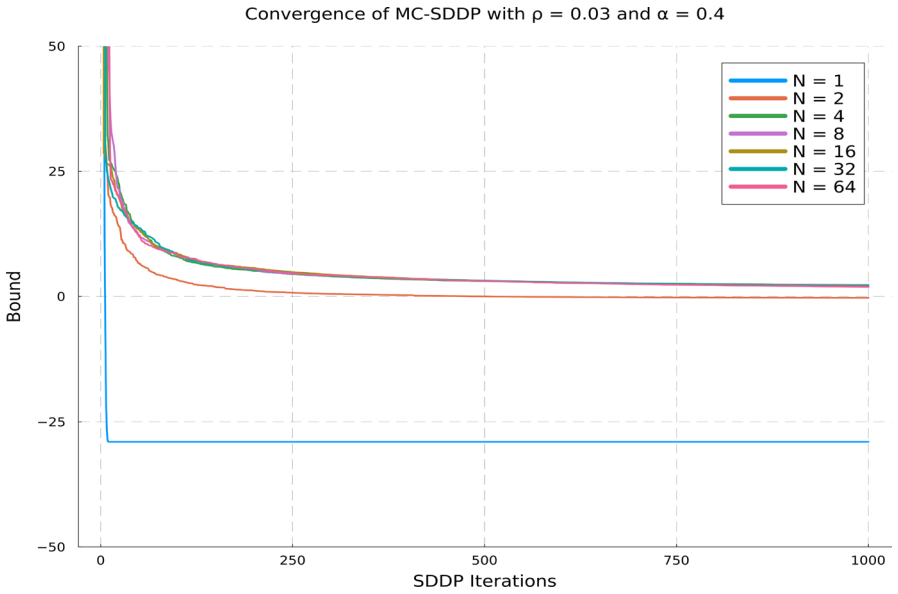


Fig. 3 Convergence of SDDP as we increase the number of quadrature points in the discretization

5 Computational results

Section 5.1 describes the data used in the specification of the model presented in Section 2. Section 5.2 presents the computational results obtained with the SDDP algorithm for the storage management problem (3). In particular, we study the accuracy and computational times obtained when the number N of quadrature points is increased. Section 5.3 builds on the numerical optimization and calculates indifference prices of different specifications of energy storage systems.

5.1 Data

In the numerical studies below, we consider the problem of battery management over one day in the intraday market. We make hourly decisions, so the number of decision stages is $T = 24$. We set the leakage rate to $l = 0$ as there is practically no leakage in a 24-hour period considered here. Furthermore, we set the discharging efficiency to $c^- = 1.05$, the charging efficiency to $c^+ = 0.95$, and the bidask spread coefficient to $\delta = 1€$. These values are not derived from empirical data, but can be easily adjusted without affecting the computations. Throughout this section, we assume that the maximum charging and discharging speeds are equal, and we express them as a fraction α of the battery capacity. In other words, $\overline{U} = \underline{U} = \alpha \overline{X}^e$.

We discretize the one-dimensional Markov process ξ from Sect. 2.4 using the conditional quadratures described in Sect. 4.2. We choose the sampling densities ϕ_t to be Gaussian with zero mean and variance σ^2. The corresponding measures are then approximated by Gauss-Hermite quadrature. For simplicity, we use the same number N of quadrature points in each stage t. We solve the resulting discretized

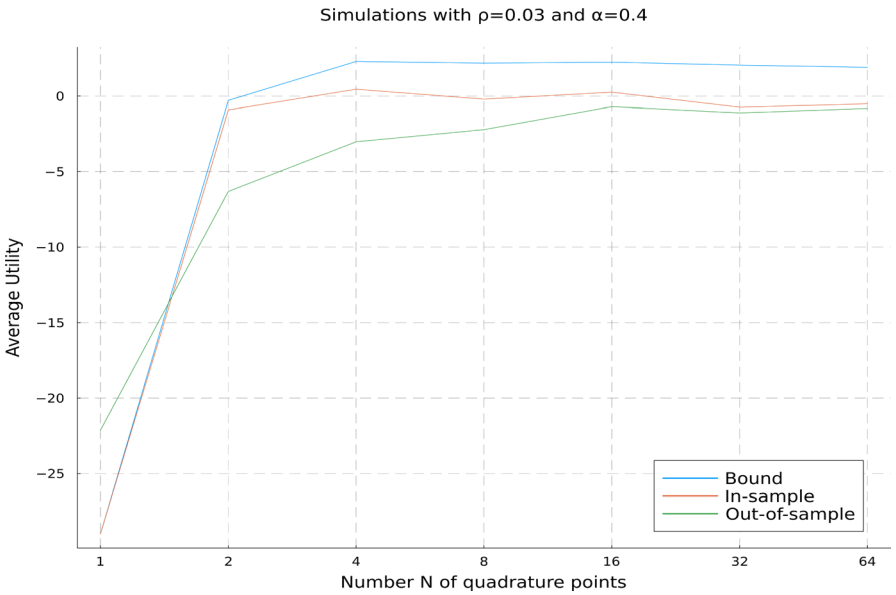


Fig. 4 Optimum value estimates for an increasing number of quadrature points. The red and blue values are given by SDDP, while the green values are obtained with out of sample simulation using the optimized Bellman functions

Bellman equations with the Julia package `SDDP.jl` Dowson and Kapelevich (2021) which implements the Markov chain SDDP algorithm outlined in Sect. 4.3. Figure 2 illustrates the discretization of the log-price over 15 days with $N = 3$ quadrature points per time.

5.2 Optimization of storage operation

We study the performance of SDDP as we increase the number N of quadrature points per period. For each N, we run 1000 iterations of SDDP. The risk aversion parameter ρ is set to 0.03 and the capacity to $\bar{X}_e = 1$MWh. In the convergence analysis of this subsection, we will assume that $\alpha := \overline{U}/\overline{X}^e = 0.4$.

Table 2 gives the computation times for 1000 iterations of the SDDP algorithm for increasing the number N of quadrature points per stage. The computation time increases roughly linearly with N.

Each backward pass of the SDDP algorithm gives an upper bound for the optimum value in terms of the piecewise linear approximations of the Bellman functions. The convergence of the upper bounds is depicted in Fig. 3 for varying number N of grid points per stage. In the deterministic case where $N = 1$, the algorithm converges in less than 10 iterations. The convergence becomes slower when N increases, but for $N \geq 4$, the upper bounds almost coincide.

In order to evaluate the quality of the cost-to-go functions provided by the SDDP algorithm, we perform an out-of-sample evaluation along 10000 scenarios randomly

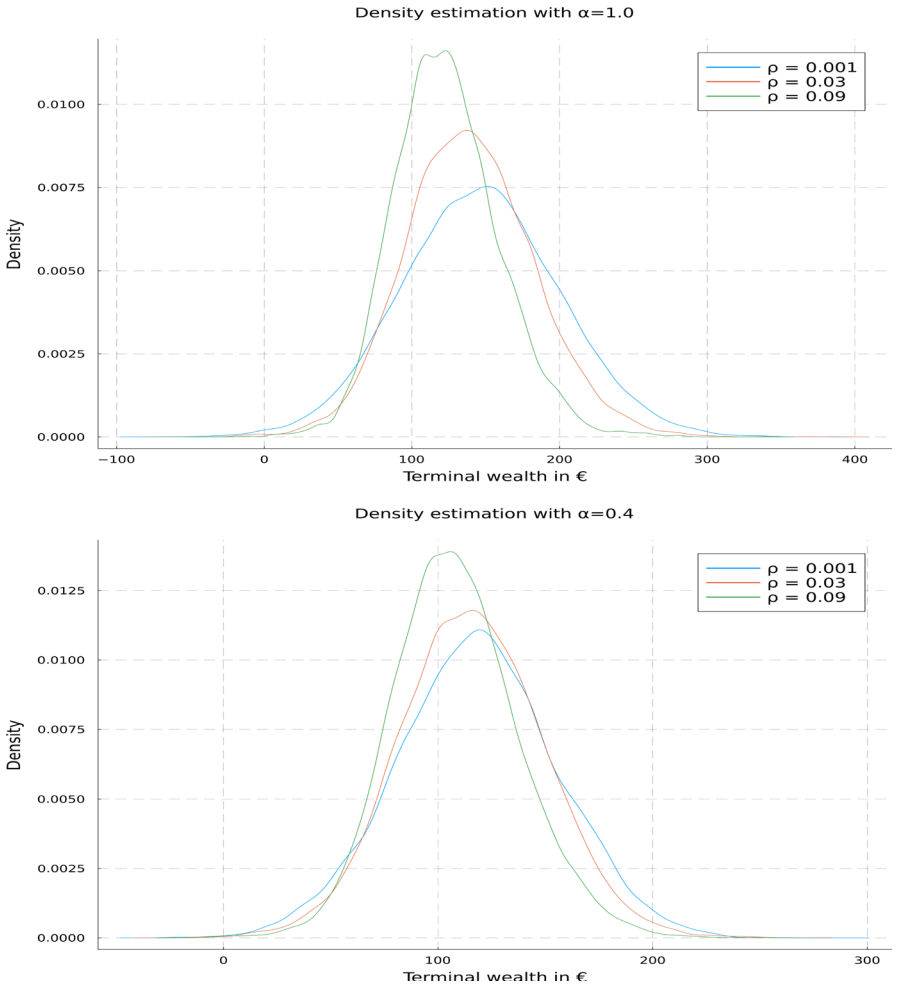


Fig. 5 Kernel density plots of the terminal wealth in out-of-sample simulations using 10000 scenarios for different values of the risk aversion parameter ρ in the exponential utility function (6). The charging speed is set to $\alpha = 0.4$ for the top figure, and to $\alpha = 1$ for the bottom figure. The capacity is set to $\bar{X}_e = 1$MWh

drawn from the continuously distributed Markov process described in Sect. 2.4. In each scenario, we implement the control obtained as in the forward pass of the SDDP algorithm as described in Sect. 4.3. However, since the random out-of-sample values of ξ_t will not coincide with the Markov states s_t^i in the finite-state approximation, we use the control obtained in the state s_t^i nearest to the sampled value of ξ_t.

Figure 4 plots the out-of-sample estimates of the objective values together with the upper bounds found after 1000 iterations of the SDDP algorithm. The figure also plots "in-sample" averages obtained by evaluating the objective along scenarios randomly drawn from the finite-state Markov chain approximation of the continuously distributed Markov process ξ. The out-of-sample averages increase and tend towards

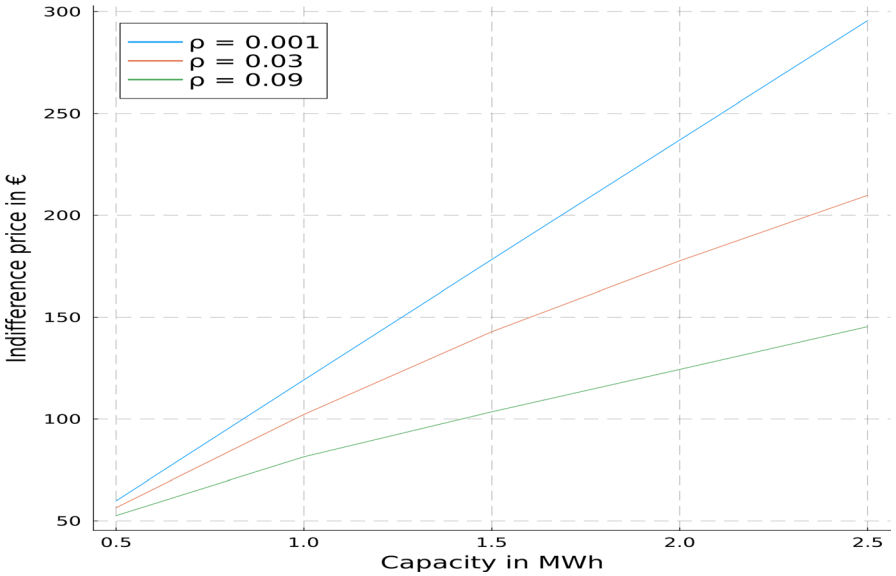


Fig. 6 Indifference prices (in euros) of storage systems with different storage capacities. Different colors correspond to agents with different risk aversion parameters. Prices are given in euros and capacity in MWh. The charging speed is set to $\alpha = 0.4$

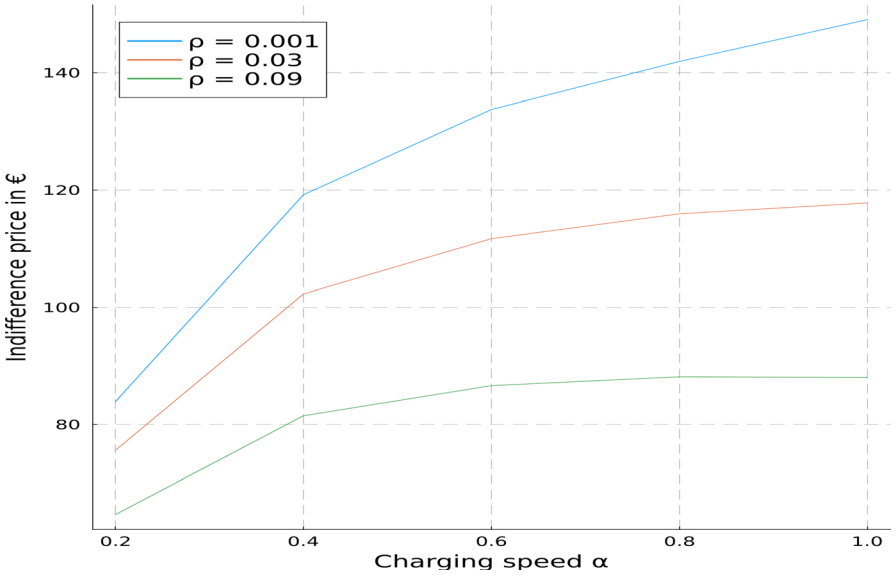


Fig. 7 Indifference prices (in euros) of storage systems with different charging speeds. The horizontal axis gives the maximum hourly change in stored energy in units of the total storage capacity. Different colors correspond to agents with different risk aversion parameters. The capacity is set to $\bar{X}_m = 1$ MWh

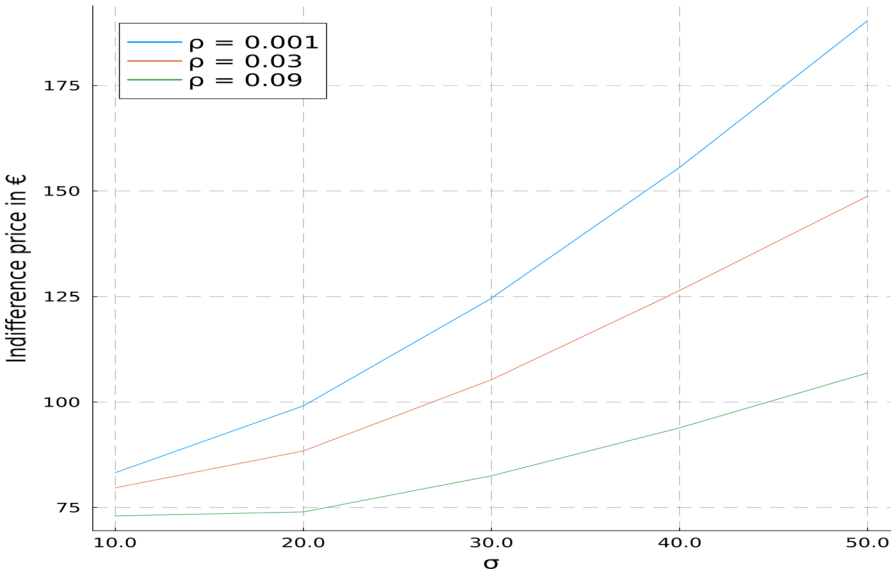


Fig. 8 Indifference prices (in euros) obtained with different values of the volatility parameter σ in the AR process (8). Different colors correspond to different risk aversions. The charging speed is set to $\alpha = 0.4$ and the capacity to $\bar{X}_m = 1$MWh

the SDDP upper bound as we increase N. This is expected, as the Markov discretization is an approximation of the reference AR-1 process. However, for $N > 8$, the improvements seem marginal.

5.3 Valuation of energy storage systems

As observed in Sect. 5.2, the SDDP algorithm gives fairly accurate results already with $N = 8$ quadrature points per period. From now on, we fix $N = 8$ and proceed to compute indifference prices of storage using the numerical evaluations of the optimum value function of (3) as described in Sect. 2.3. Since we are working with the exponential utility function, the indifference prices are given by Proposition 1 after the solution of just one instance of the battery management problem. This section studies the dependence of the optimal solutions and the indifference prices when we vary the parameters of the battery and the trader's risk aversion.

Figure 5 displays kernel-density plots of the terminal wealth obtained with strategies optimized for different risk aversion. As expected, higher risk aversion results in a thinner left tail of the terminal wealth distribution. The plots the kernel density estimates for two values of the charging speed. The effect of the risk aversion seems to influence the terminal wealth more for batteries with a higher charging speed. A highly risk-averse agent will tend to operate a battery more sparingly due to the market price risk. A less risk-averse agent, on the other hand, tends to take full advantage of a high-speed battery.

Figure 6 plots indifference prices for batteries with different capacities and agents with different risk aversions. As one might expect, the valuation of a storage increases

with the storage capacity. With higher risk aversion, however, the valuations seem to saturate for larger capacities. This is quite natural as a highly risk-averse agent will avoid large positions in stored electricity due to the price risk. Figure 7 plots the indifference prices for storage systems with different charging speeds. Again, the valuations are more sensitive for less risk-averse agents. Finally, Fig. 7 gives indifference prices for varying values of the volatility parameter σ in the electricity price model (8). The volatility can be thought of as the agent's uncertainty concerning the evolution of electricity prices. For all levels of risk aversion, higher volatility makes the storage more valuable to the agent (Fig. 8).

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